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TEST # 03

SYLLABUS

Do not open this Test Booklet until you are ready with below stated syllabus.

PHYSICS:

Thermodynamics, Kinetic Theory Of Gases, Oscillations And Waves, Experimental Skills : Simple Pendulum-Dissipation Of Energy By Plotting A Graph Between, The Square Of Amplitude And Time, Metre Scale - The Mass Of A Given Object By The Principle Of Moments, Speed Of Sound In Air At Room Temperature Using A Resonance Tube, Specific Heat Capacity Of A Given (I) Solid And (II) Liquid By Method Of Mixtures.

CHEMISTRY:

Some Basic Principles of Organic Chemistry, Hydrocarbons, Purification and Characterisation of Organic Compounds.

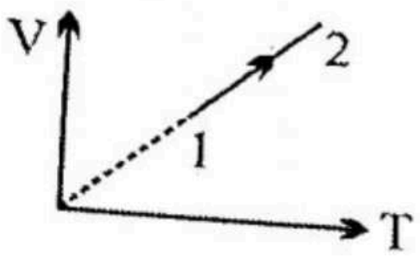
BIOLOGY:

Photosynthesis in Higher Plants, Plant Growth and Development, Respiration in Plants, Neural Control And Co-ordination (Nervous System), Chemical Co-ordination And Integration (Endocrine System), Locomotion And Movement (Muscles, Skeletal System)

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1. An ideal gas undergoes the process 1 $\longrightarrow$ 2 as shown in the figure, the heat supplied and work done in the process is ΔQ and ΔW respectively. The ratio $\Delta Q : \Delta W$ is



(1) $\gamma : \gamma - 1$

(2) γ

(3) $\gamma - 1$

(4) $\gamma - 1 : \gamma$

2. ***Assertion :*** When a bottle of cold carbonated drink is opened, a slight fog forms around the opening.
Reason : Adiabatic expansion of the gas causes lowering of temperature which start condensation of water vapours.

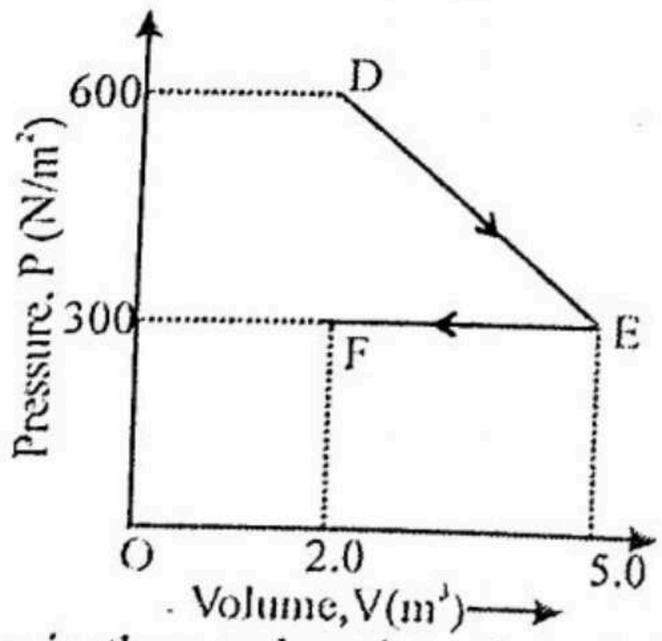
(1) Both Assertion and Reason are true but Reason is not the correct explanation of Assertion.

(2) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.

(3) Assertion is true but Reason is false.

(4) Assertion is false but Reason is true.

3. A thermodynamic system is taken from an original state (D) to an intermediate state (E) by the linear process shown in figure:



Its volume is then reduced to the original value from E to F by an isobaric process. Calculate the total work done by the gas from D to E and E to F

(1) 450 J

(2) 1350 J

(3) – 1350 J

(4) 900 J

4. Molar specific heat of water is ?

(1) 2R

(2) 3R

(3) 9R

(4) R

5. A carnot's engine works between a source at a temperature of 27°C, and a sink at –123°C. Its efficiency is :-

(1) 0.5

(2) 0.25

(3) 0.75

(4) 0.4

6. Oxygen and hydrogen are at the same temperature T. The ratio of the mean kinetic energy of oxygen molecules to that of the hydrogen molecules will be:

(1) 16 : 1

(2) 1 : 1

(3) 4 : 1

(4) 1 : 4

7. The temperature of an ideal gas is increased from 27°C to 927°C. The rms speed of its molecules becomes:

(1) Twice

(2) Half

(3) Four times

(4) One-fourth

8. A gas at 300 K is enclosed in container. Now container is placed on fast moving train while train is in motion, the temperature of gas :

(1) rise above 300 K

(2) fall below 300 K

(3) remain unchanged

(4) become unsteady

9. If the pressure of an ideal gas contained in a closed vessel is increased by 0.5%, the increase in temperature is 2 K. The initial temperature of the gas is :

(1) 27°C

(2) 127°C

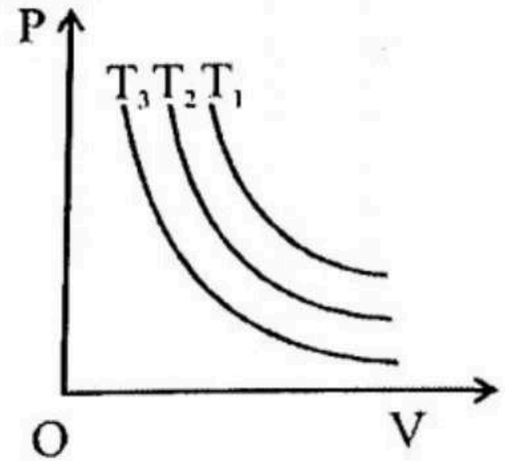
(3) 300°C

(4) 400°C

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10. The isothermal diagram of a gas at three different temperature T_1, T_2 and T_3 is shown in the given figure, Then :-



(1) $T_1 < T_2 < T_3$ (2) $T_1 < T_2 > T_3$

(3) $T_1 > T_2 > T_3$ (4) $T_1 > T_2 < T_3$

11. 2 mole of an ideal monoatomic gas at temperature T_0 are mixed with 4 mole of another ideal monoatomic gas at temperature $2T_0$, then the temperature of mixture is

(1) $\frac{5}{3}T_0$ (2) $\frac{3}{2}T_0$

(3) $\frac{4}{3}T_0$ (4) $\frac{5}{4}T_0$

12. For O_2 gas, $C_p - C_v = a$ and for N_2 gas; $C_p - C_v = b$, if C_p & C_v are gram specific heat then ratio of a to b (a : b) is -

(1) $\frac{7}{8}$ (2) $\frac{8}{7}$

(3) $\frac{1}{1}$ (4) None of these

13. If a metallic sphere gets cooled from 62°C to 50°C in 10 minutes and in the next 10 minutes gets cooled to 42°C, then the temperature of the surroundings is :

(1) 30°C (2) 36°C

(3) 26°C (4) 20°C

14. A black body has maximum wavelength λ_m at 2000 K. Its corresponding wavelength at 3000 K will be:

(1) $\frac{16}{81}\lambda_m$ (2) $\frac{81}{16}\lambda_m$

(3) $\frac{3}{2}\lambda_m$ (4) $\frac{2}{3}\lambda_m$

15. A spring elongated by length 'ℓ' when a mass 'M' is suspended to it. Now a tiny mass 'm' is attached and then released, its time period of oscillation is:-

(1) $2\pi\sqrt{\frac{(M+m)\ell}{Mg}}$

(2) $2\pi\sqrt{\frac{m\ell}{Mg}}$

(3) $2\pi\sqrt{\ell/g}$

(4) $2\pi\sqrt{\frac{M\ell}{(m+M)g}}$

16. The time period of a mass suspended from a spring is 5 sec. The spring is cut into four equal parts and the same mass is now suspended from one of its parts. The period is now :-

(1) 5 s (2) 2.5 s

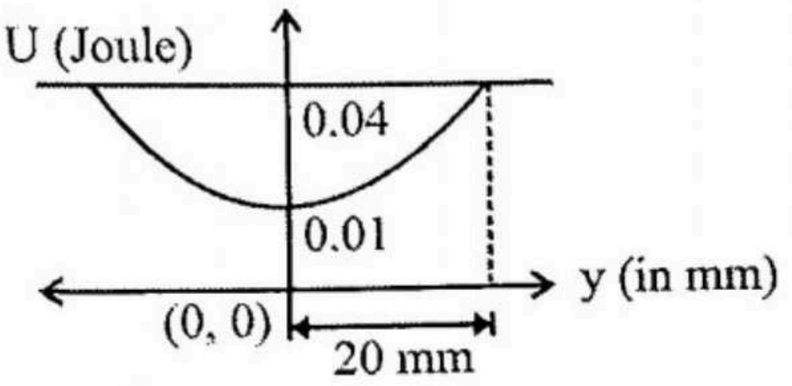
(3) 1.25 s (4) (5/16) s

17. A simple pendulum performs simple harmonic motion about x = 0 with an amplitude a and time period T. The speed of the pendulum at $x = \frac{a}{\sqrt{2}}$ will be :-

(1) $\frac{3\pi a}{T}$ (2) $\frac{4\pi a}{T}$

(3) $\frac{\sqrt{2}\pi a}{T}$ (4) $\frac{aT}{\pi^2}$

18. The variation of potential energy of harmonic oscillator is as shown in figure. The spring constant is



(1) 1×10^2 N/m

(2) 150 N/m

(3) 0.667×10^2 N/m

(4) 3×10^2 N/m

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19. Two waves are represented by : $y_1 = 4 \sin 404\ \pi t$ and $y_2 = 3 \sin 400\ \pi t$. Then :

(1) beat frequency is 4 Hz and the ratio of maximum to minimum intensity if 49 : 1

(2) beat frequency is 2 Hz and the ratio of maximum to minimum intensity is 49 : 1

(3) beat frequency is 2 Hz and the ratio of maximum to minimum intensity is 1 : 49

(4) beat frequency is 4 Hz and the ratio of maximum to minimum intensity is 1 : 49

20. A sonometer wire supports 4 kg load and vibrates in fundamental mode with tunning fork of frequency 416 Hz. The length of wire between the bridges is now doubled. In order to maintain fundamental mode the load should be :-

(1) 1 kg (2) 2 kg (3) 8 kg (4) 16 kg

21. When a wave travels in a medium, the particle displacement is given by the equation $y = a \sin 2\pi(bt - cx)$ where a, b and c are constants. The maximum particle velocity will be twice the wave velocity if :-

(1) $c = \frac{1}{\pi a}$ (2) $c = \pi a$

(3) $b = ac$ (4) $b = \frac{1}{ac}$

22. A string of length 3 m and linear mass density 0.0025 kg/m is fixed at both ends. One of its resonance frequency is 252 Hz. The next higher resonance frequency is 336 Hz. Then the fundamental frequency will be -

(1) 84 Hz (2) 63 Hz

(3) 126 Hz (4) 168 Hz

23. In a resonance tube experiment, the first resonance is obtained for 10 cm of air column and the second for 32 cm. The end correction for this apparatus is–

(1) 0.5 cm (2) 1.0 cm

(3) 1.5 cm (4) 2 cm

24. A student is performing experiment with resonance tube apparatus in Physics lab to find the speed of sound at room temperature. He got first two resonating lengths of air column as 17 cm and 51 cm, using tuning fork of frequency 512 Hz. Find speed of sound at room temperature.

(1) 348 m/s (2) 358 m/s

(3) 332 m/s (4) 322 m/s

25. 100 g of ice at 0°C is mixed with 100 g of water at 100°C. What will be the final temperature of the mixture ?

(1) 10°C (2) 20°C

(3) 30°C (4) 40°C

26. Steam at 100°C is added slowly to 1400 gm of water at 16°C until the temperature of water is raised to 80°C. The mass of steam required :-

(1) 120 gm (2) 240 gm

(3) 160 gm (4) 320 gm

27. The ratio of the speed of sound in nitrogen gas to that in helium gas at 300 K is :

(1) $\sqrt{\frac{2}{7}}$ (2) $\sqrt{\frac{1}{7}}$

(3) $\frac{\sqrt{3}}{5}$ (4) $\frac{\sqrt{6}}{5}$

28. The period of a simple pendulum inside a stationary lift is T. The lift accelerates upwards with an acceleration of $\frac{g}{3}$. The time period of pendulum will be :-

(1) $\sqrt{2}T$ (2) $\frac{T}{\sqrt{2}}$ (3) $\frac{\sqrt{3}}{2}T$ (4) $\frac{T}{3}$

29. A pendulum is hung from the roof of a sufficiently high building and is moving freely to and fro like a simple harmonic oscillator. The acceleration of the bob of the pendulum is 20 m/s^2 at a distance of 5 m from the mean position. The time period of oscillation is :-

(1) 2π s (2) π s (3) 2 s (4) 1 s

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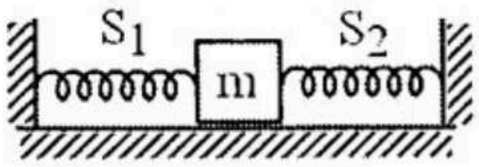
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30. A particle executes simple harmonic motion along a straight line with an amplitude A. The potential energy is maximum when the displacement is :

(1) $\pm A$ (2) Zero

(3) $\pm \frac{A}{2}$ (4) $\pm \frac{A}{\sqrt{2}}$

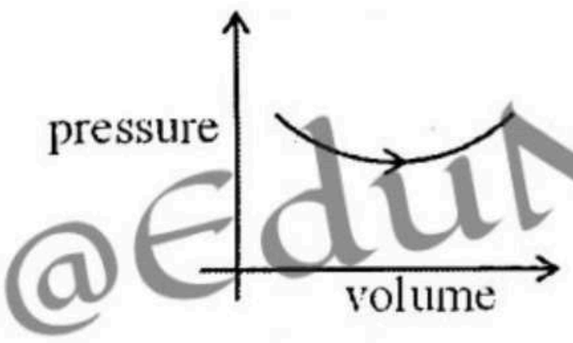
31. When a mass m is connected individually to two springs S_1 and S_2, the oscillation frequencies are V_1 and V_2, if the same mass is attached to the two springs in given figure, the oscillation frequency would be :



(1) $V_1 + V_2$ (2) $\sqrt{V_1^2 + V_2^2}$

(3) $\left(\frac{1}{V_1} + \frac{1}{V_2}\right)^{-1}$ (4) $\sqrt{V_1^2 - V_2^2}$

32. Consider the process on a system shown in figure below. During the process, the rate with which the work is being done by the system :



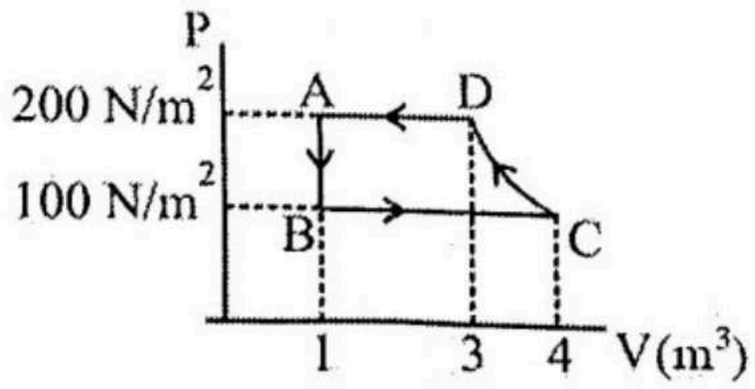
(1) Continuously increases

(2) Continuously decreases

(3) First increases then decreases

(4) First decreases then increases

33. The P-V diagram of a diatomic ideal gas system going under cyclic process is as shown. The work done during an adiabatic process CD is ($\gamma = 1.4$)



(1) −500 J (2) −400 J

(3) 400 J (4) 200 J

34. A thermally insulated vessel contain an ideal gas of molar mass M and ratio of specific heats γ. It is moving with speed v and is suddenly brought to rest. Assuming no heat is lost to the surroundings. Its temperature increased by :-

(1) $\frac{\gamma - 1}{2(\gamma - 1)R} MV^2$

(2) $\frac{(\gamma - 1)}{2\gamma R} MV^2$

(3) $\frac{\gamma MV^2}{2R}$

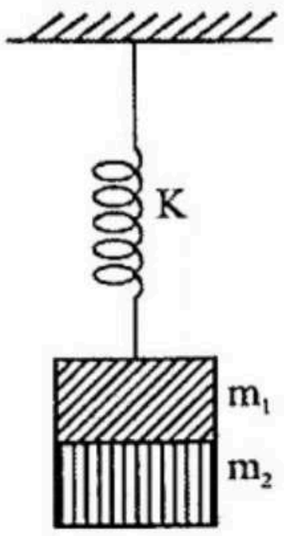
(4) $\frac{(\gamma - 1)}{2R} MV^2$

35. A container of volume $1m^3$ is divided into two equal compartments by a partition. One of these compartments contains an ideal gas at 300 *K*. The other compartment is vacuum. The whole system is thermally isolated from its surroundings. The partition is removed and the gas expands to occupy the whole volume of the container. Its temperature now would be :

(1) 300 K (2) 239 K

(3) 200 K (4) 100 K

36. Two masses m_1 and m_2 are suspended together by a massless spring of constant k. When the masses are in equilibrium, m_1 is removed without disturbing the system; the amplitude of vibration is:



(1) $m_1 g/k$

(2) $m_2 g/k$

(3) $\frac{(m_1 + m_2)g}{k}$

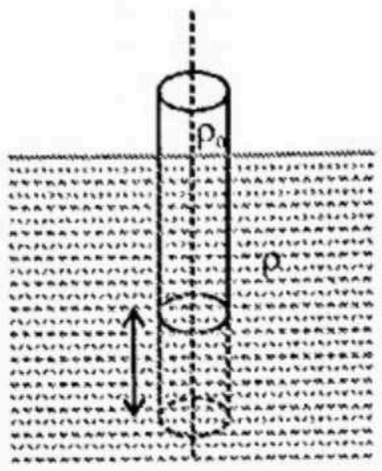
(4) $\frac{(m_2 - m_1)g}{k}$

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37. A solid cylinder of density ρ_0, cross-section area A and length ℓ floats in a liquid $\rho(>\rho_0)$ with its axis vertical, as shown. If it is slightly displaced downward and released, the time period will be :-



(1) $2\pi\sqrt{\frac{\ell}{g}}$ (2) $2\pi\sqrt{\frac{\rho_0\ell}{\rho g}}$

(3) $2\pi\sqrt{\frac{\rho\ell}{\rho_0 g}}$ (4) $2\pi\sqrt{\frac{\ell}{2g}}$

38. The amplitude of a particle executing SHM is 4 cm. At the mean position the speed of the particle is 16 cm/sec. The distance of the particle from the mean position at which the speed of the particle becomes $8\sqrt{3}$cm/s, will be :-

(1) $2\sqrt{3}$cm (2) $\sqrt{3}$ cm

(3) 1cm (4) 2cm

39. A simple pendulum has a time period T in vacuum. Its time period when it is completely immersed in a liquid of density one-eight of the density of material of the bob is :-

(1) $\sqrt{\frac{7}{8}}T$ (2) $\sqrt{\frac{5}{8}}T$

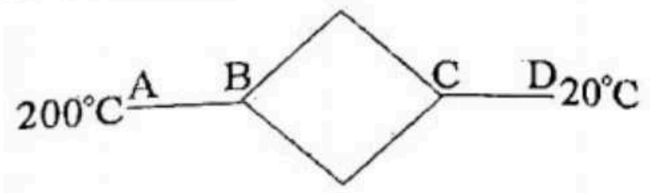
(3) $\sqrt{\frac{3}{8}}T$ (4) $\sqrt{\frac{8}{7}}T$

40. An open organ pipe of length ℓ is sounded together with another open organ pipe of length $\ell + x$ in their fundamental tones. Speed of sound in air is v. The beat frequency heard will be : $(x << \ell)$:-

(1) $\frac{vx}{4\ell^2}$ (2) $\frac{v\ell^2}{2x}$

(3) $\frac{vx}{2\ell^2}$ (4) $\frac{vx^2}{2\ell}$

41. Six identical conducting rods are joined as shown in figure. Points A and D are maintained at temperature 200°C and 20°C respectively. The temperature of the junction B will be :-



(1) 120°C (2) 100°C

(3) 140°C (4) 80°C

42. An organ pipe filled with a gas at 27°C resonates at 400 Hz in its fundamental mode. If it is filled with the same gas at 90°C, the resonance frequency at the same mode will be :-

(1) 420 Hz (2) 440 Hz

(3) 484 Hz (4) 512 Hz

43. A source of frequency 2400 Hz when vibrated over the mouth of a closed organ pipe is in unison at 300 K. The beats produced per second when temperature rises by 1 K is :

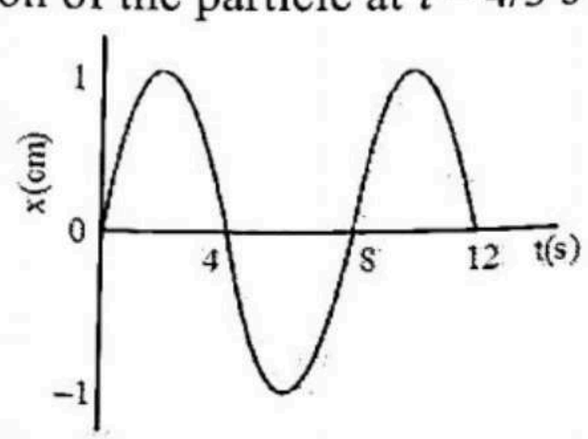
(1) 8 (2) 6 (3) 4 (4) 2

44. If two tuning forks A and B are sounded together, they produce 4 beats per second. A is then slightly loaded with wax, they produce same beats when sounded again. The initial frequency of A is 256 Hz. The frequency of B will be :-

(1) 250 Hz (2) 252 Hz

(3) 260 Hz (4) 262 Hz

45. The x-t graph of a particle undergoing simple harmonic motion is shown below. The acceleration of the particle at $t = 4/3$ s is



(1) $\frac{\sqrt{3}}{32}\pi^2$ cm/s^2 (2) $\frac{-\pi^2}{32}$ cm/s^2

(3) $\frac{\pi^2}{32}$ cm/s^2 (4) $-\frac{\sqrt{3}}{32}\pi^2$ cm/s^2

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SUBJECT : CHEMISTRY

46. Which of the following homologous series has incorrect general formula :-

(1) Alkyne C_nH_{2n-2}

(2) Alkanol $C_nH_{2n+2}O$

(3) Alkanal $C_nH_{2n+1}O$

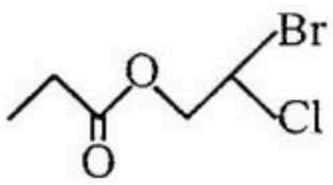
(4) Carboxylic acid $C_nH_{2n}O_2$

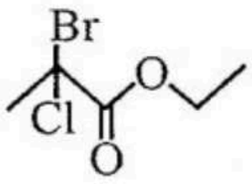
47. Minimum number of carbon atoms present in an ester are :-

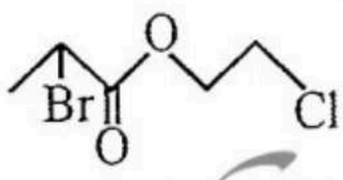
(1) 2 (2) 1

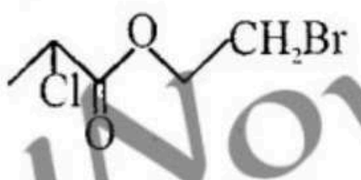
(3) 4 (4) 3

48. Structure of the compound Ethyl 2-bromo-2-chloropropanoate is :-

(1) 

(2) 

(3) 

(4) 

49.

Mixture		Purification technique	
(A)	Chloroform and Aniline	(P)	Distillation
(B)	Glycerol and spentlye	(Q)	Chromatography
(C)	Aniline and water	(R)	Steam Distillation
(D)	Amino acid	(S)	Distillation under Reduced pressure

Correct Match:-

(1) A-P, B-Q, C-R, D-S

(2) A-P, B-S, C-R, D-Q

(3) A-P, B-R, C-S, D-Q

(4) A-R, B-S, C-P, D-Q

50. (I) $CH_3-CH_2-NH-CHO$ (II) $CH_3-\underset{\underset{NH_2}{|}}{CH}-CHO$

Which type of isomerism is observed between I & II :

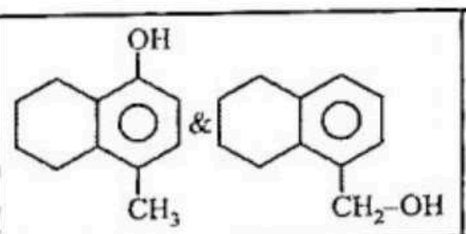
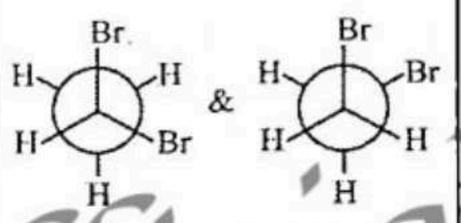
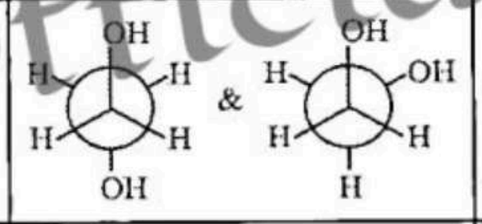
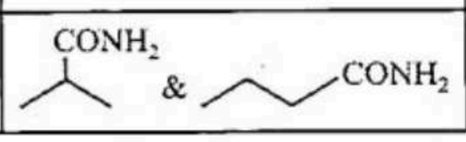
(1) Chain isomerism

(2) Position isomerism

(3) Functional isomerism

(4) Metamerism

51. Match the column :

(A)		(P)	Position isomer
(B)		(Q)	Functional group isomer
(C)		(R)	Chain isomer
(D)		(S)	Conformer

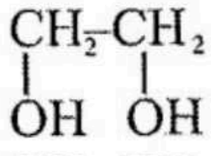
(1) A-P,B-Q,C-R,D-S

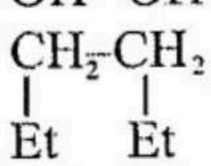
(2) A-Q, B-P, C-R, D-S

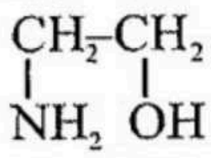
(3) A-Q,B-P,C-S,D-R

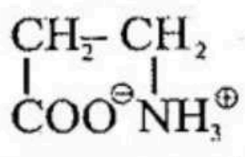
(4) A-S,B-P,C-S,D-P

52. In which of the following gauche form will be not most stable form :-

(1) 

(2) 

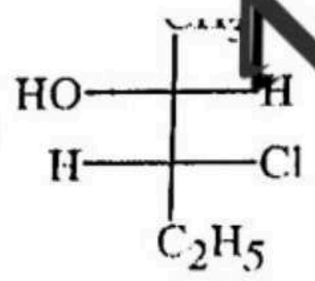
(3) 

(4) 

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53. The full name of compound  is :-

(1) (2R, 3R)- 3-chloro-2-pentanol

(2) (2R, 3S)- 3-chloro-2-pentanol

(3) (2S, 3R)- 3-chloro-2-pentanol

(4) (2S, 3S)- 3-chloro-2-pentanol

54. Which one of the following series contains electrophiles only ?

(1) H_2O, SO_3, H_3O^+ (2) NH_3, H_2O, $AlCl_3$

(3) $AlCl_3$, SO_3, $\overset{+}{N}O_2$ (4) H_2O, $\overset{+}{Cl}$, NH_3

55. Which of the following is the correct order for –I effect ?

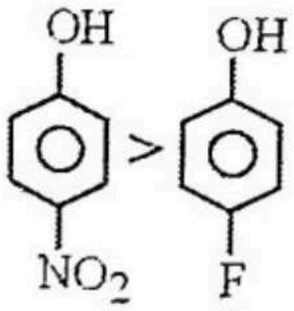
(1) $-NO_2 > -CN > -F > -C_6H_5$

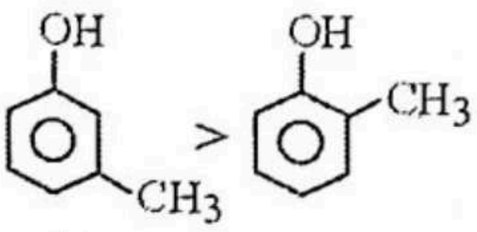
(2) $-C_6H_5 > -F > -CN > -NO_2$

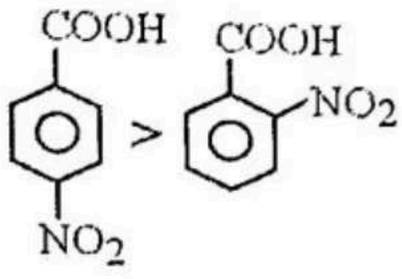
(3) $-F > -CN > -C_6H_5 > -NO_2$

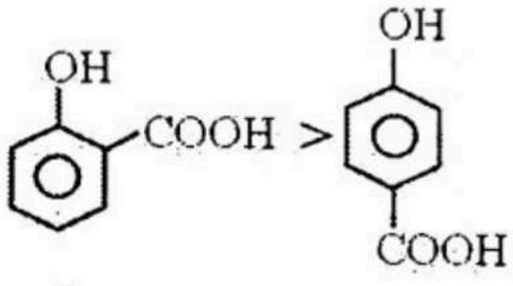
(4) $-CN > C_6H_5 > -NO_2 > -F$

56. Which of the following show incorrect relation?

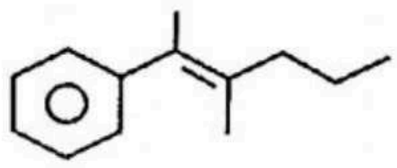
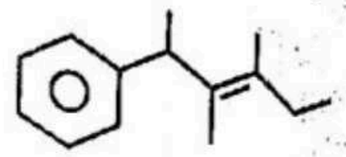
(1)  ⇒ Acidic strength order

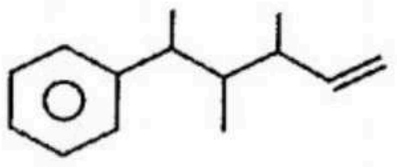
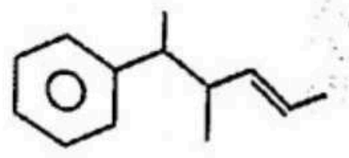
(2)  ⇒ Acidic strength order

(3)  ⇒ Acidic strength order

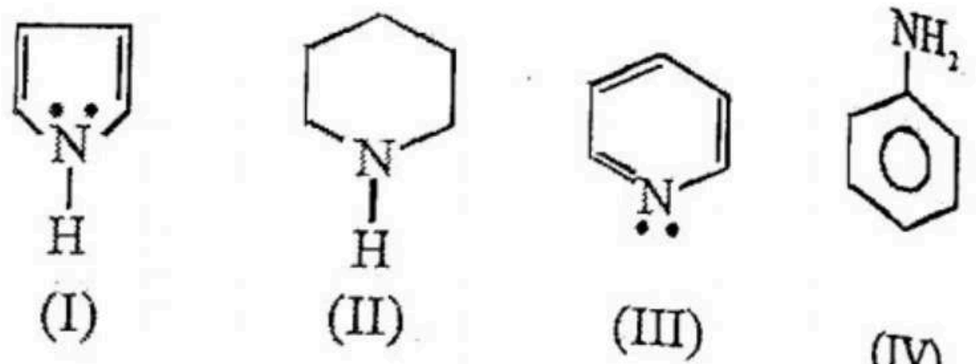
(4)  ⇒ Acidic strength order

Which of the following has least heat hydrogenation ?

(1)  (2) 

(3)  (4) 

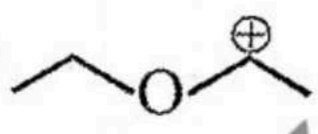
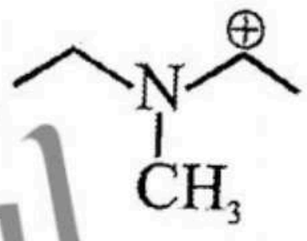
58. Correct decreasing order of basic strength fo following:-



(I) (II) (III) (IV)

(1) I > II > III > IV (2) II > III > I > IV

(3) II > IV > I > III (4) II > III > IV > I

59. (a)  (b) 

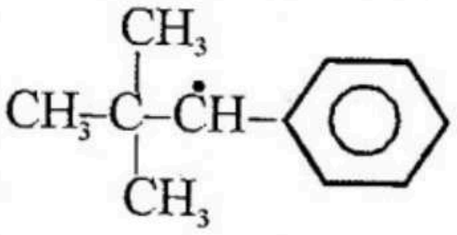
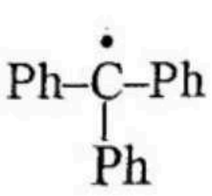
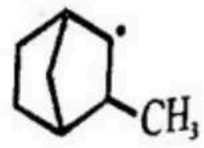
(c) 

Which is correct order for stability of above carbocations :-

(1) c > a > b (2) b > a > c

(3) c > b > a (4) b > c > a

60. Hyperconjugation occurs in :-

(i) (ii) (iii)

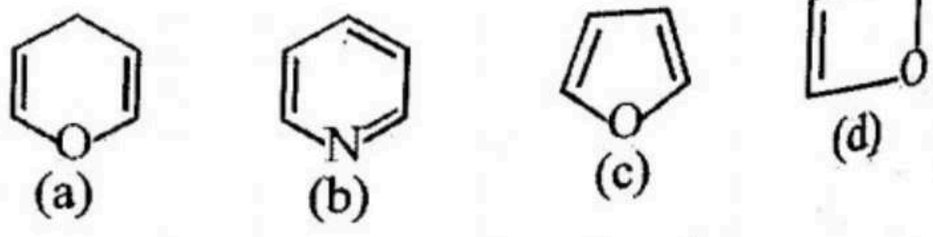
(1) Only (i)

(2) (i) & (ii)

(3) (i), (ii) & (iii)

(4) only (iii)

61. Which of the following is aromatic :-



(a) (b) (c) (d)

(1) a and b (2) b and c

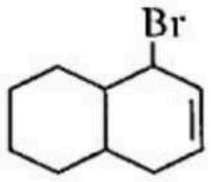
(3) c and d (4) a and d

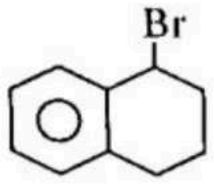
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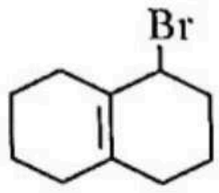
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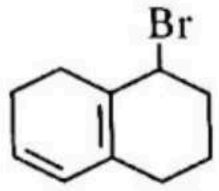
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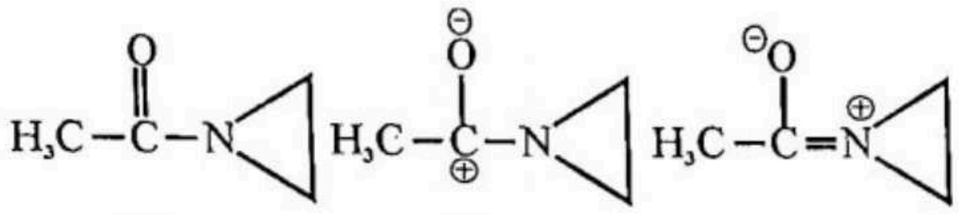
62. which of the following is maximum reactive towards De-hydrohalogination

(1) 

(2) 

(3) 

(4) 

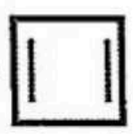
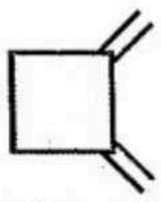
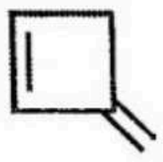
63. 

(X) (Y) (Z)

The correct stability order of the given canonical structure is :

(1) X > Y > Z (2) Z > X > Y

(3) X > Z > Y (4) Y > Z > X

64. (I)  (II)  (III) 

Correct statement about these three compounds :-

(1) Stability order : II > III > I

(2) Stability order : III > I > II

(3) Heat of combustion : I > III > II

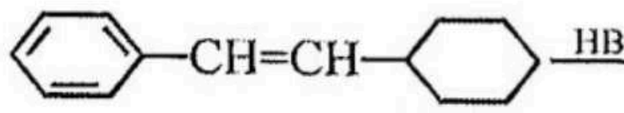
(4) Both (1) and (3) statements are correct

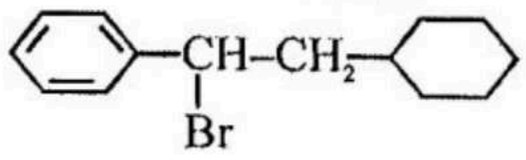
65. The correct order of boiling point of following compounds is :

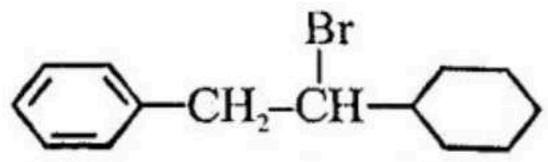
A) n-pentane B) Isopentane C) Neopentane

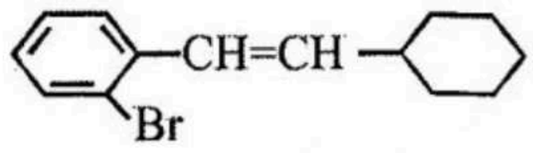
(1) A > B > C (2) B > A > C

(3) C > B > A (4) A > C > B

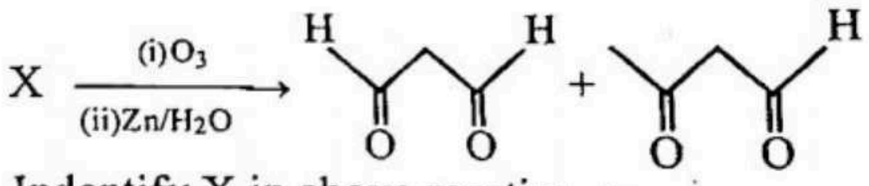
66.  ? (Major product)

(1) 

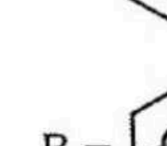
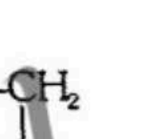
(2) 

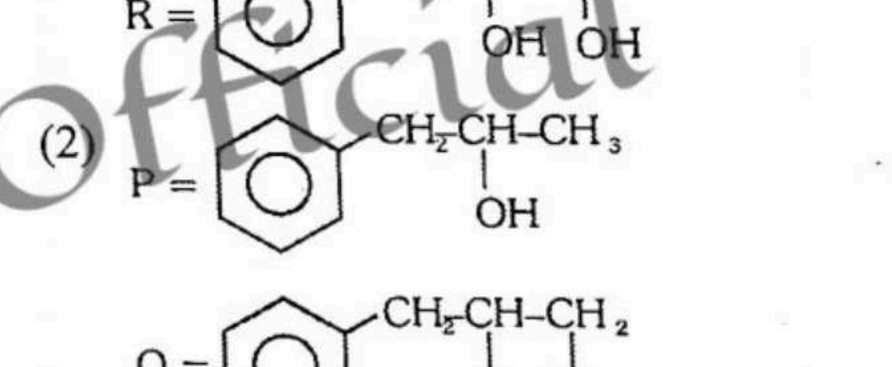
(3) 

(4) Both (1) & (2)

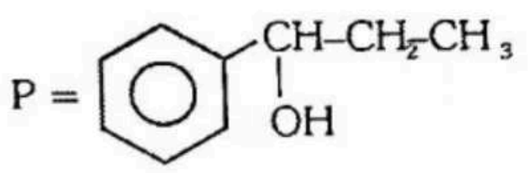
67. X 

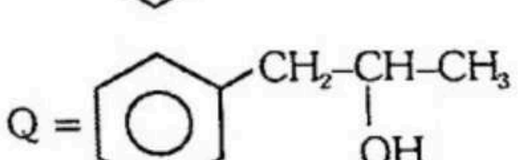
Indentify X in above reaction .:-

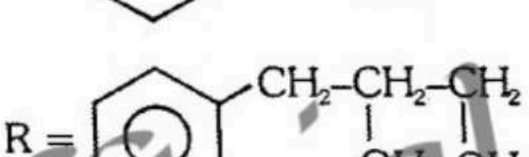
(1)  (2) (3)  (4)

68. 

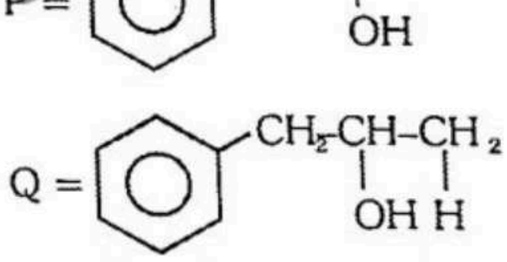
Products P, Q and R are respectively –

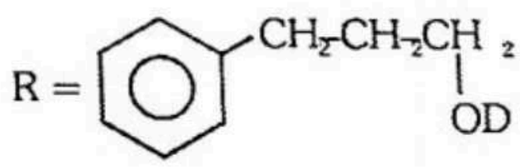
(1) 

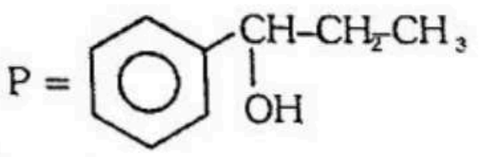


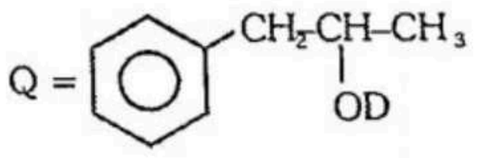


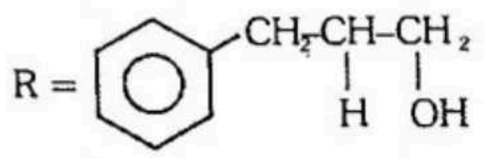
(2)

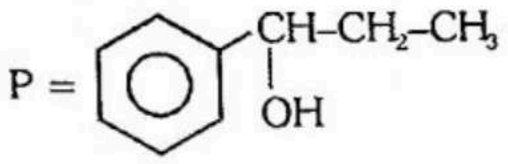


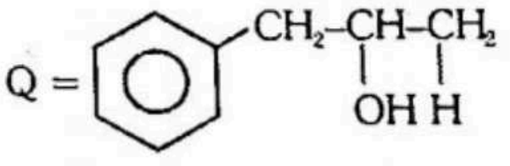


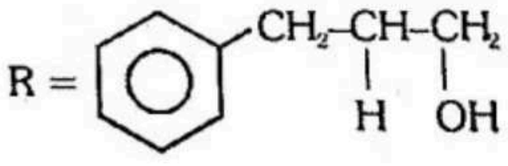
(3) 





(4) 



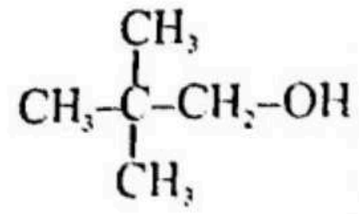


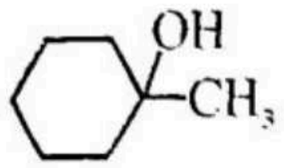
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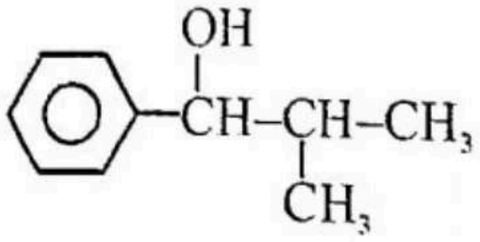
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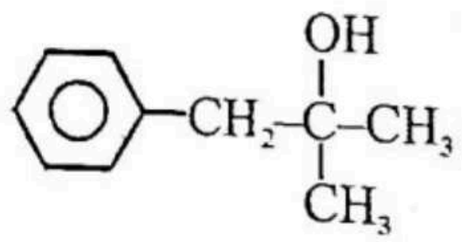
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69. Which of the following alcohol undergoes dehydration reaction most easily :

(1) 

(2) 

(3) 

(4) 

70. Reactivity of halogen in halogenation:-

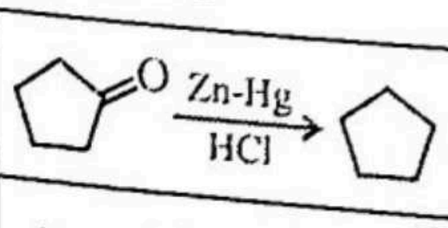
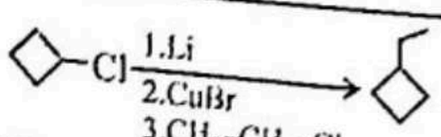
(1) $F_2 < Cl_2 < Br_2 < I_2$

(2) $Cl_2 > F_2 > Br_2 > I_2$

(3) $F_2 > Cl_2 > Br_2 > I_2$

(4) $Br_2 > Cl_2 > F_2 > I_2$

71. Match the following columns -

	Column-I (Reaction)		Column-II (Name of reaction)
A.	$CH_3-CH_2-Br \xrightarrow[\text{Dry ether}]{Na}$ n-Butane	P	Clemmensen's reduction
B.	$\begin{matrix}CH_2-COONa\\ \\ CH_2-COONa\end{matrix} \xrightarrow[\text{Electrolysis}]{NaOH}$ Ethene	Q	Correy House synthesis
C.		R	Kolbe's electrolysis
D.		S	Wurtz reaction

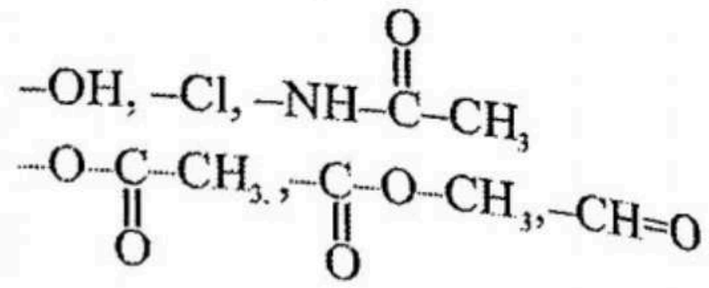
(1) A-S, B-R, C-Q, D-P

(2) A-R, B-S, C-P, D-Q

(3) A-Q, B-R, C-P, D-S

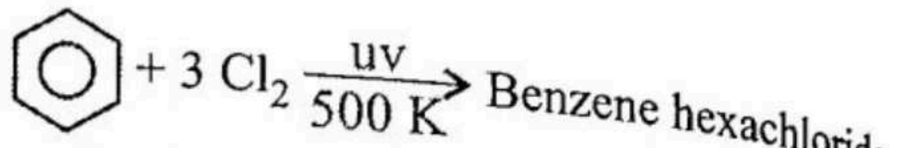
(4) A-S, B-R, C-P, D-Q

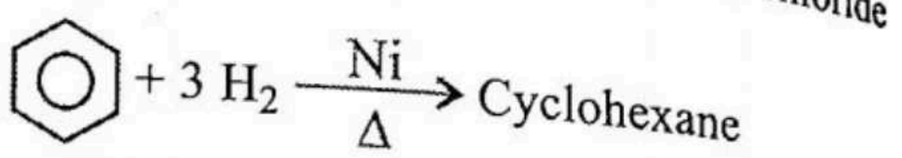
72. How many of following groups are ortho-para directing?

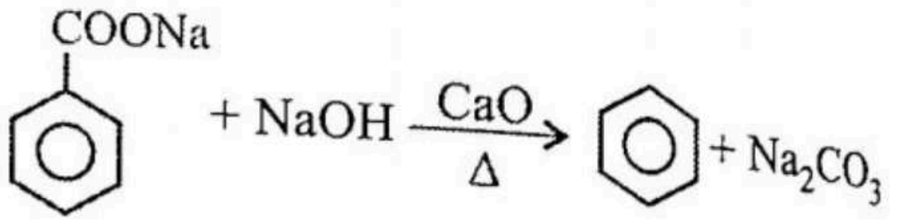


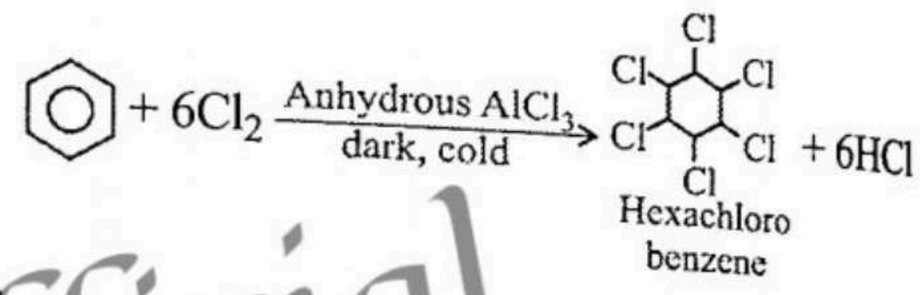
(1) 4 (2) 6 (3) 5 (4) 3

73. Which reaction is incorrect ?

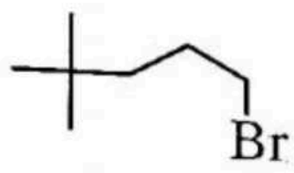
(1) 

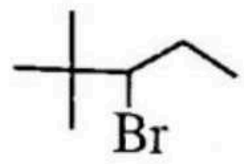
(2) 

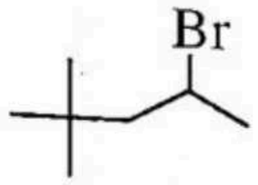
(3) 

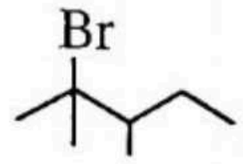
(4) 

74. 

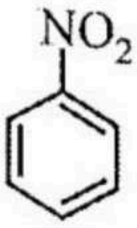
(1) 

(2) 

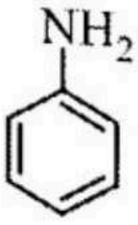
(3) 

(4) 

75. In which of the following compound, Kjeldahl method is applicable :

(1) 

(2) 

(3) 

(4) Ph–N=N–Ph

76. Mixture of Anthracene and benzoic Acid can be separated by

(1) Fractional distillation

(2) Sublimation

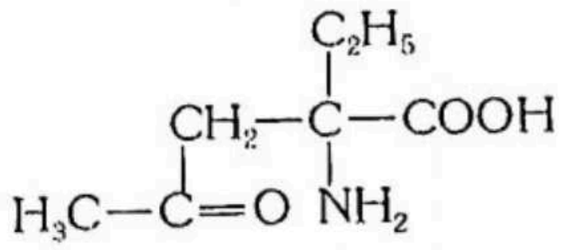
(3) Chromatography

(4) Filteration

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77. The IUPAC name of given compound is :-



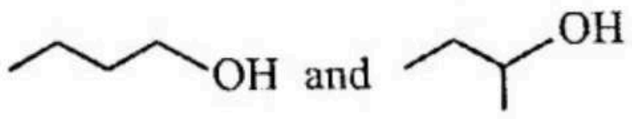
(1) 2–Oxo–4–amino–4–ethyl pentanoic acid

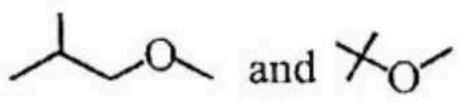
(2) 2–Amino–4–oxo–2–ethyl pentanoic acid

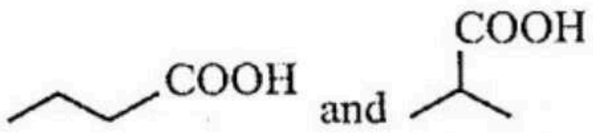
(3) 3–Amino–3–carboxy–5–hexanone

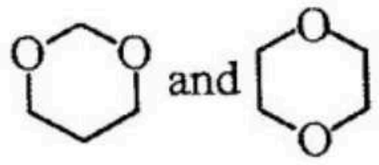
(4) 2–Amino–2–ethyl–4–oxo pentanoic acid

78. Which of the following pairs of compounds are not position isomer :-

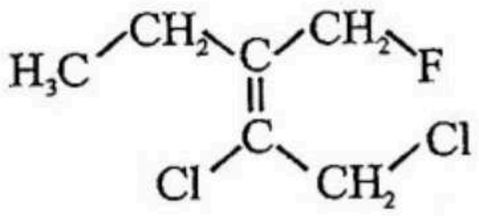
(1) 

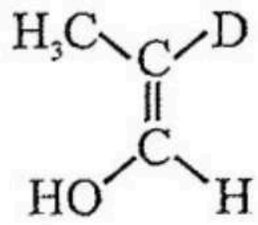
(2) 

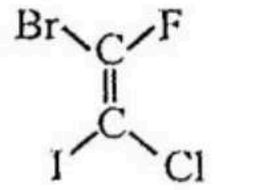
(3) 

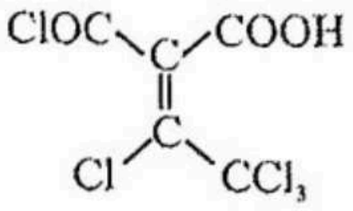
(4) 

79. Which of the following shows E-configuration ?

(1) 

(2) 

(3) 

(4) 

80. **Statement-I** : Molecules having chiral carbons will always be optically active.

Statement-II : it is not necessary to have chiral carbon to show chirality.

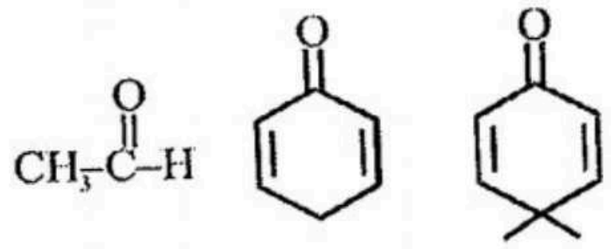
(1) Both statement-I and statement-II are correct

(2) Both statement-I and Statement-II are incorrect

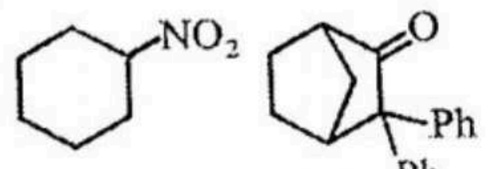
(3) Statement-I is incorrect and statement-II is correct

(4) Statement-I is correct and statement-II is incorrect

81. Consider following molecules :



(A) (B) (C)



(D) (E)

Which can show tautomerism :

(1) A, B, C

(2) B, C, E

(3) A, B, D

(4) B, C, E, F

82. Match the List-I with List-II :-

	List - I (Compounds)		**List - II (pK_a- Value)**
(a)	o-Nitrophenol	(i)	10
(b)	p-Nitrophenol	(ii)	8.3
(c)	m-Nitrophenol	(iii)	7.2
(d)	Phenol	(iv)	7.1

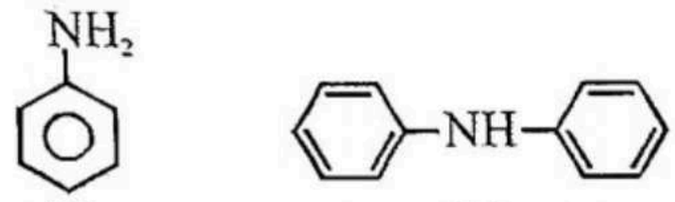
(1) a-iv, b-iii, c-ii, d-i

(2) a-iv, b-ii, c-iii, d-i

(3) a-iii, b-iv, c-i, d-ii

(4) a-iii, b-iv, c-ii, d-i

83. Arrange the following according to basicity :-



(P) (Q)



(R) (S)

(1) P > S > R > Q

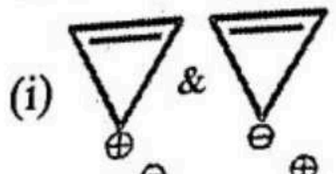
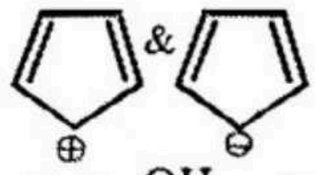
(2) S > R > P > Q

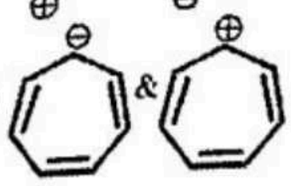
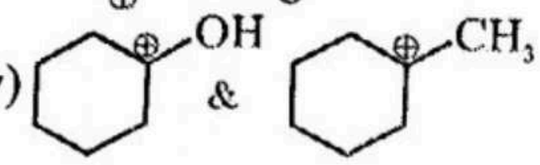
(3) S > Q > R > P

(4) S > R > Q > P

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84. In which pair second ion is more stable than first?

(i)  (ii) 

(iii)  (iv) 

(1) (i) and (ii)

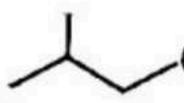
(2) (ii) and (iii)

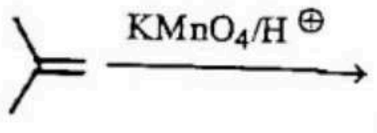
(3) (ii) and (iv)

(4) (iii) and (iv)

85. Which of following reaction is correct :

(1) $Ph-\underset{\underset{CH_3}{|}}{C}=CH-CH_3 \xrightarrow[H_2O_2]{HBr} Ph-\underset{\underset{CH_3}{|}}{\overset{\overset{Br_3}{|}}{C}}-CH_2-CH_3$

(2) $(CH_3)_3CH \xrightarrow[\text{Oxidation}]{KMnO_4(O)}$  (1°Alc.)

(3)  $+ CO_2 + H_2O$

(4) $CH_3-C\equiv CH \xrightarrow{Hg^{2+}/dil.H_2SO_4} CH_3-CH_2-CH=O$

6. The correct statement regarding the basicity of arylamines is :-

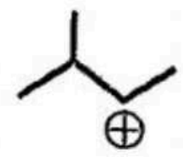
(1) Arylamines are generally less basic than alkylamines because the nitrogen lone-pair electrons are delocalized by interaction with the aromatic ring π electron system.

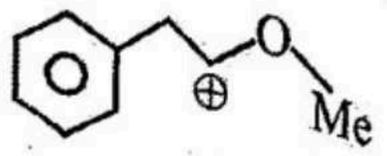
(2) Arylamines are generally more basic than alkylamines because the nitrogen lone-pair electrons are not delocalized by interaction with the aromatic ring π electron system.

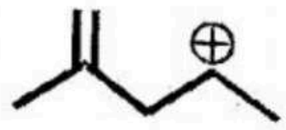
(3) Arylamines are generally more basic than alkylamines because of aryl group.

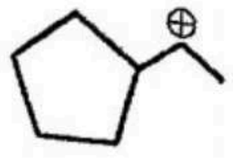
(4) Arylamines are generally more basic than alkylamines, because the nitrongen atom in arylamines is sp-hybridized.

87. Carbocation rearrangement is not observed in

(1) 

(2) 

(3) 

(4) 

88. Which of the following alcohols on dehydration with conc. H_2SO_4 at 440 K will give propylene ?

(i) *n*-propyl alcohol

(ii) Isobutyl alcohol

(iii) Isopropyl alcohol

(iv) *n*-butyl alcohol

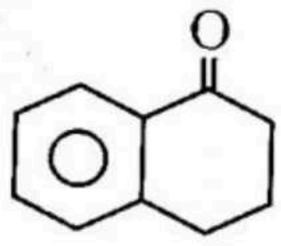
(1) (ii), (iii)

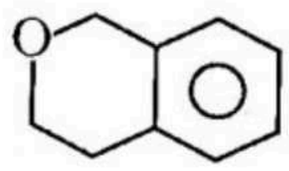
(2) (i), (iii)

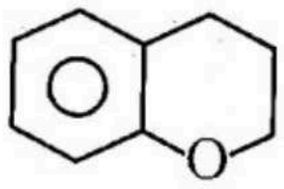
(3) (i), (ii), (iii)

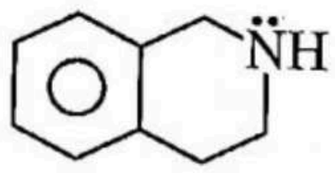
(4) (ii), (iii), (iv)

89. Which is maximum reactive towards electrophilic substitution reaction (ESR)

(1) 

(2) 

(3) 

(4) 

90. 0.5 gm of an organic compound gave 50 ml of N_2 collected at 300 K temperature and 715 mm pressure By DUMAS method. Calculate percentage of nitrogen in the compound (Aqueous Tension at 300 K = 15 m.m)

(1) 17.465

(2) 15.899

(3) 12.475

(4) 10.475

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91. Which of the following statement is incorrect ?

(1) Splitting of water molecule take place on the inner side of membrane of thylakoid.

(2) Primary electron acceptor (Pheophytin) is located towards the outer side of thylakoid membrane.

(3) ATP synthase enzyme consist of two part one called CF_0 is embedded in the membrane and form a transmembrane channel.

(4) Cyclic photophosphorylation involves both PSI and PSII.

92. Match the following:

Column-I		Column-II	
(a)	Primary acceptor of CO_2 in C_3 plant	(i)	PEP
(b)	Primary acceptor of CO_2 in C_4 plant	(ii)	PGA
(c)	First CO_2 fixation product in C_3 plant	(iii)	OAA
(d)	First CO_2 fixation product in C_4 plant	(iv)	RUBP

(1) (a)-(i), (b)-(iii), (c)-(ii), (d)-(iv)

(2) (a)-(iii), (b)-(ii), (c)-(i), (d)-(iv)

(3) (a)-(iv), (b)-(i), (c)-(ii), (d)-(iii)

(4) (a)-(i), (b)-(iv), (c)-(iii), (d)-(ii)

93. **Statement-I :** C_4 plants lack photorespiration.

Statement-II : C_4 plants have a mechanism that increases the concentration of CO_2 at the site of RuBisCO enzyme.

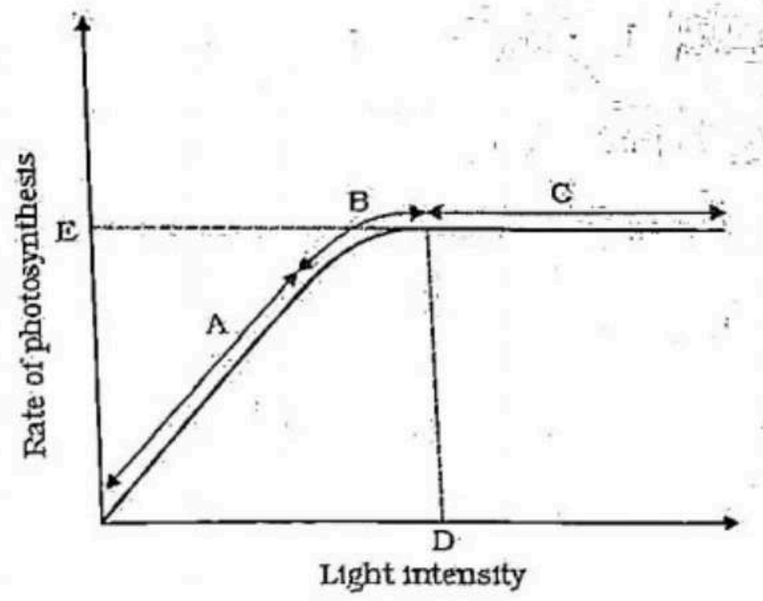
(1) Both **Statement I** and **Statement II** are incorrect.

(2) Both **Statement I** and **Statement II** are correct.

(3) **Statement I** is correct but **Statement II** is incorrect.

(4) **Statement I** is incorrect but **Statement II** is correct

94.



What does the "B" label represent :

(1) Light limiting factor

(2) Saturation point of light

(3) Factor other than CO_2 are limiting

(4) Factor other than light are limiting

95. Where exactly enzymes of dark reaction are present within chloroplast ?

(1) Thylakoid membrane

(2) Grana

(3) Stromal lamella

(4) Stroma of chloroplast

96. What is the main factor that limits the rate of photosynthesis on a clear day ?

(1) CO_2

(2) Water

(3) Light

(4) Chlorophyll

97. Which of the following organism was used in experiment of T.W. Engelmann ?

(1) *Cladophora* (2) Mint

(3) *Albugo* (4) *Chlorella*

98. Kranz anatomy is found in -

(1) C_3 plants (2) C_4 plants

(3) CAM plants (4) All plants

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99. Calvin cycle occurs in -

(1) C_3 plants only

(2) C_4 plants only

(3) CAM plants only

(4) In all plants

100. Number of ATP and $NADPH.H^+$ required respectively in regeneration step for synthesizing one molecule of sucrose.

(1) 12 ATP, 6 $NADPH.H^+$

(2) 24 ATP, 12 $NADPH.H^+$

(3) 6 ATP, 0 $NADPH.H^+$

(4) 12 ATP, 0 $NADPH.H^+$

101. RuBisCO show :-

(1) Only carboxylase activity

(2) Only oxygenase activity

(3) Both (1) and (2)

(4) Hydrolase activity

102. Match the following.

Column-I		Column-II	
(a)	Complex-I	(i)	Succinate dehydrogenase
(b)	Complex-II	(ii)	NADH dehydrogenase
(c)	Complex-IV	(iii)	Cytochrome bc_1
(d)	Complex-III	(iv)	Cytochrome C oxidase

	(a)	(b)	(c)	(d)
(1)	i	ii	iii	iv
(2)	ii	i	iv	iii
(3)	iii	i	ii	iv
(4)	iv	iii	ii	i

103. Which of the following complex contains cytochrome a, a_3 and two copper centres ?

(1) Complex I

(2) Complex II

(3) Complex III

(4) Complex IV

104. **Assertion** : Krebs cycle occurs in mitochondrial matrix.

Reason : All the enzyme of TCA cycle present in membrane of mitochondria.

(1) Both **Assertion** and **Reason** are true but **Reason** is NOT the correct explanation of **Assertion**.

(2) **Assertion** is true but **Reason** is false.

(3) **Assertion** is false but **Reason** is true.

(4) Both **Assertion** and **Reason** are true and **Reason** is the correct explanation of **Assertion**.

105. Which of the following is the site of link reaction ?

(1) Cytoplasm

(2) Mitochondria

(3) Chloroplast

(4) Both (1) & (2)

106. Which one of the following is not the cofactor of link reaction ?

(1) Mg^{2+} (2) CoA

(3) NAD^+ (4) Ca^{+2}

107. While the oxygen concentration is low, the product of glycolysis pyruvate is turned into _____.

(1) Acetyl CoA

(2) Ethanol and CO_2

(3) Citric acid

(4) Phosphoenol pyruvate

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108. The continued oxidation of acetyl Co-A via the TCA cycle requires the continued replenishment of _____ :

(1) Malic acid

(2) Succinic acid

(3) Oxaloacetic acid

(4) α-ketoglutaric acid

109. Fill in the blank space with the correct option.

_____can be used as respiratory substrates in plants.

(1) Only carbohydrates.

(2) Only proteins and fats.

(3) Only organic acids.

(4) Carbohydrates, proteins, fats and organic acids.

110. The breaking of C-C bonds of complex compounds leading to release of considerable amount of energy is called :-

(1) Photosynthesis

(2) Respiration

(3) Translocation

(4) Transpiration

111. Which of the following statement is not correct for plants.

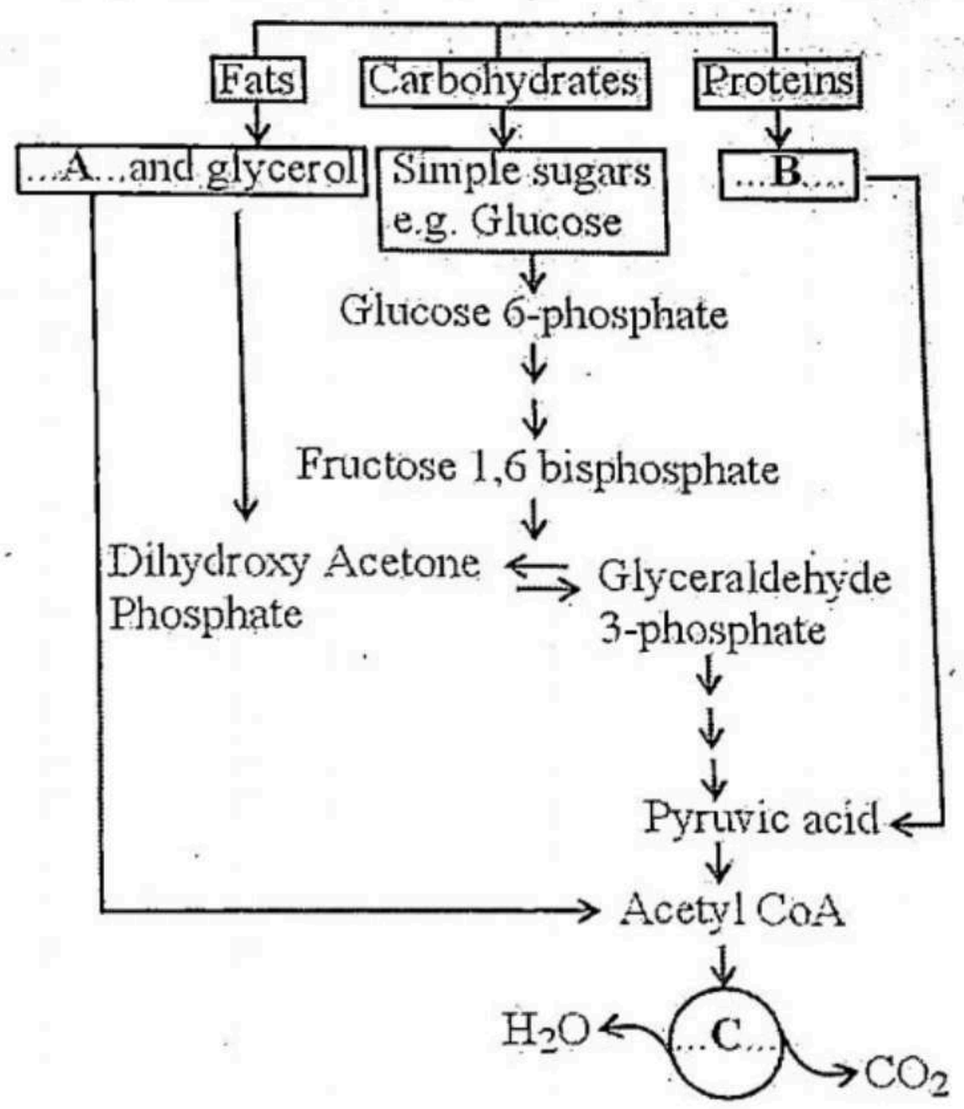
(1) There is very large transport of gases from one plant part to another.

(2) Plants can get along without respiratory organs.

(3) Plants respire at rates for lower than animal do.

(4) the distance that gases must diffuse even in large bulky plants is not great.

112. Identify A, B and C :-



(1) A → Lipid, B → Fatty acid, C → Link reaction

(2) A → Fatty acids, B → Amino acids, C → Kreb's cycle

(3) A → Monosaccharide, B → Sugar, C → Glycolysis

(4) A → Fatty acid, B → Lipid, C → ETS

113. **Assertion :-** Plant growth is unique and localised.

Reason :- Plants retain the capacity for unlimited growth throughout their life and that growth occur at specific locations.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Assertion is False but the Reason is True.

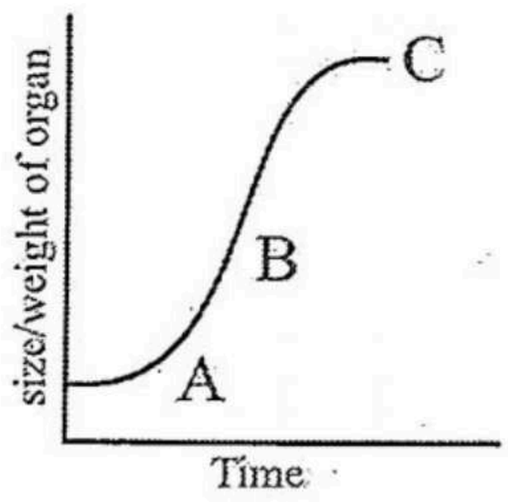
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114. Match the following

	Match - I		Match - II
(i)	Terpenes	(a)	Abscisic acid
(ii)	Adenine derivatives	(b)	Gibberellin
(iii)	Derivatives of carotenoids	(c)	Ethylene
(iv)	Gas	(d)	Cytokinine

	(i)	(ii)	(iii)	(iv)
(1)	a	b	c	d
(2)	b	a	c	d
(3)	b	d	a	c
(4)	c	d	a	b

115. Which of the following is correct regarding this diagram :-



(1) Arithmetic growth

A = Lag phase, B= Exponential phase

C = Stationary phase

(2) Geometric growth

A=Exponential phase, B=Lag phase

C=Stationary phase

(3) Geometric growth

A=lag phase, B=Exponential phase

C=Stationary phase

(4) Relative growth

A=lag phase, B=Exponential phase

C=Stationary phase

116. Which of the following statement is incorrect ?

(1) Cytokinin helps overcome the apical dominance.

(2) Auxin promotes flowering in pineapples

(3) Ethylene also controls xylem differentiation and helps in cell division.

(4) Gibberellin is used to increase the length of grapes stalk.

117. Which of the following hormones induce rooting on stem cuttings ?

(1) Auxin (2) Ethylene

(3) Cytokinin (4) Gibberellins

118. Phytohormone that promotes bolting in many plants with rosette habit is-

(1) Auxin (2) Cytokinins

(3) Gibberellins (4) Abscisic acid

119. The discovery of gibberellins is associated with a disease of :-

(1) Wheat (2) Rice

(3) Maize (4) Barley

120. Retention of chlorophyll and delayed senescence of leaves is done by

(1) Auxin (2) GA_3

(3) Cytokinins (4) Ethylene

121. __A__ Conferred in 1910, the release of a volatile substance __B__ from ripened oranges that hastened the ripening of stored unripened bananas :-

Identify A and B.

(1) (A)-F. skoog, (B)-Auxin

(2) (A)-E. kurosawa, (B)-Gibberellic acid

(3) (A)-H. H. Cousins, (B)-Ethylene

(4) (A)-Miller et. al., (B)-Ethylene

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122. Plant hormone that plays an important role in seed development, maturation and dormancy is-

(1) Auxin

(2) Ethylene

(3) Cytokinin

(4) Abscissic acid

123. Find correct statement -

I. In green plants H_2O is hydrogen donor and is oxidised to O_2.

II. Jan Ingenhousz hypothesised that plants restore to the air whatever animal and candle remove.

III. Engelmann provided evidence for production of glucose when plants grow.

(1) I & II (2) II & III

(3) II only (4) I only

124. **Assertion** : Except sciophytes and plants of dense forest, light is rarely a limiting factor in nature.

Reason : Light saturation occurs at 10 percent of full summer sun light.

(1) Both assertion and reason are true and the reason is a correct explanation of the assertion

(2) Both assertion and reason are true and the reason is not a correct explanation of the assertion

(3) Assertion is true but reason is false

(4) Both assertion and reason are false

125. Choose correct option for fill in the blanks-

In half leaf experiment cotton was soaked with __'A'__ and set up was placed in __'B'__.

(1) A–NaCl, B–Light

(2) A–KOH, B–Dark

(3) A–NaOH, B–Dark

(4) A–KOH, B–Light

126. Law of limiting factors was given by :-

(1) Blackman (2) Hill

(3) Calvin (4) Van Niel

127. Fill in the blanks-

_____ also receives reducing equivalents via _____ that is generated during oxidation of succinate.

Select the correct option-

(1) $NADH_2$, Ubiquinone

(2) Ubiquinone, $FADH_2$

(3) Plastoquinone, $FADH_2$

(4) Ubiquinone, $NADH_2$

128. Enzyme of TCA cycle which is competitively inhibited by malonate.

(1) Malate dehydrogenase

(2) Isocitrate dehydrogenase

(3) Fumarase

(4) Succinate dehydrogenase

129. Yeasts poison themselves to death when the concentration of alcohol reaches about _____ .

(1) 13 % (2) 7 %

(3) 5 % (4) 2 %

130. Process in respiration that occurs in anaerobic organisms :

(1) Kreb cycle

(2) Link reaction

(3) Glycolysis

(4) Electron transport system

131. End products of glycolysis are :-

(1) 2 ATP, 2 $FADH_2$ & 2 Pyrvic acid

(2) 2 ATP, 2 NADH & 2 Pyruvic acid

(3) 2 ATP, 2 NADPH & Pyruvic acid

(4) 8 ATP, 2 NADH & 2 Pyruvic acid

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132. Which of the following is an assumption of the respiratory balance sheet ?

(1) Respiration is a sequential, orderly pathway.

(2) Only glucose is being respired.

(3) Alternative substrates are entering in the respiratory pathway.

(4) Both 1 and 2

133. Choose the incorrect pair :-

(1) Meristematic phase - Cells are rich in protoplasm posess large conspicuous nuclei.

(2) Elongation phase - Increased vacuolation

(3) Maturation phase - Cell wall are thin and cellulosic

(4) Secondary growth - Lateral meristem.

134. Application of which hormone can delay abscission of fruits and leaves at early stage ?

(1) Ethylene (2) ABA

(3) Auxin (4) Cytokinin

135. Search for natural substances with cytokinin like activities led to the isolation of _____ from corn-kernels and coconut milk.

(1) Kinetin (2) Zeatin

(3) Indol-3-acetic acid (4) Indole butyric acid

136. Which structure is tracts of myelinated nerve fibres?

(1) Association areas (2) Hippocampus

(3) Corpus callosum (4) Amygdala

137. Respiratory centre is situated in

(1) Cerebellum (2) Medulla oblongata

(3) Hypothalamus (4) Cerebrum

138. Association area of brain is responsible for all of the following functions except

(1) Intersensory association

(2) Memory

(3) Communication

(4) Emotional and sexual behaviour

English

139. The correct sequence of meninges of brain from inside to out side in human is:

(1) Duramater – Arachnoid – Piamatar

(2) Arachnoid – Piamatar – duramatar

(3) Piamater – Arachnoid – Duramater

(4) Piamater – Duramater – arachnoid

140. Except one, all are parts of brain stem :

(1) Pons (2) Medulla

(3) Cerebellum (4) Mid brain

141. The major coordinating centre for sensory and motor signaling is–

(1) Thalamus (2) Pons

(3) Medulla (4) Optic lobes

142. The neural system provides an organized network of __A__ connection for a __B__ coordination. Choose the correct option for A, B to complete the given statement.

(1) A - point to point, B - slow

(2) A - chemical, B - fast

(3) A - point to point, B - fast

(4) A - Chemical, B - slow

143. **Assertion (A) :** Impulse transmission across an electrical synapse is always faster than that across a chemical synapse.

Reason (R) : Transmission of an impulse across chemical synapses is very similar to impulse conduction along a single axon.

(1) Both Assertion and Reason are correct but Reason is not the correct explanation of Assertion.

(2) Both Assertion and Reason are correct and Reason is the correct explanation of Assertion.

(3) Assertion is correct but Reason is incorrect.

(4) Both Assertion and Reason are incorrect.

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144. Five events in the transmission of nerve impulse across the synapse are given below

A. Opening of specific ion channels allows the entry of ions, a new action potential is generated in the post-synaptic neuron.

B. Neurotransmitter binds to the receptor on post synaptic membrane

C. Synaptic vesicle fuses with pre-synaptic membrane, neurotransmitter releases into synaptic cleft.

D. Depolarization of pre-synaptic membrane

E. Arrival of action potential at axon terminal.

In which sequence to the events occur ?

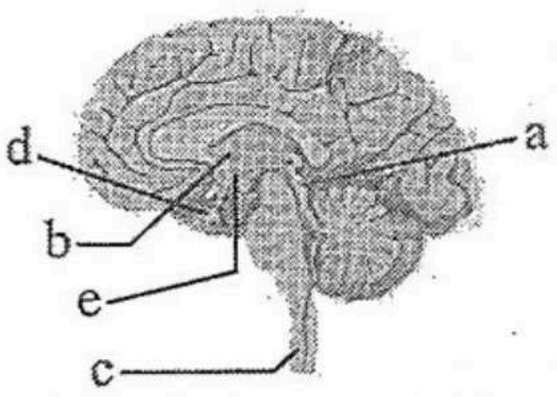
(1) E → D → C → B → A

(2) A → B → C → D → E

(3) A → B → D → C → E

(4) E → D → C → A → B

145.



Choose the incorrect statement from the following

(1) 'a' passes through mid brain

(2) 'c' contains a number of centres which control body temperature

(3) 'b' is a major coordinating centre.

(4) 'a' is a part of the mid brain.

146. Glucocorticoids stimulates :-

(A) Gluconeogenesis

(B) Lipolysis

(C) Proteolysis

(1) A only

(2) A, B and C

(3) B and C only

(4) B only

147. Which of the following is not related to hypersecretion of collip's hormone :-

(1) Osteoporosis (2) Renal stone

(3) Hypercalcaemia (4) Tetany

148. Which of the following hormone helps in maintaining the normal rhythms of sleep-wake cycle, body temperature ?

(1) MSH (2) Melatonin

(3) Thyroxin (4) (1) & (2) both

149. Adenohypophysis consists of :-

(1) Pars distalis (2) Pars intermedia

(3) Pars nervosa (4) (1) & (2) both

150. Deaf mutism & Irregular menstruation can be caused due to :-

(1) Hypersecretion of thyroxine

(2) Hyposecretion of thyroxine

(3) Hyposecretion of sex hormone

(4) Hypersecretion of sex hormone

151. Chemical nature of hormones are :-

(1) Steroid (2) Amino acid

(3) Proteins (4) All of the above

152. ICSH is another name of–

(1) TSH

(2) FSH

(3) MSH

(4) LH

153. Which of the following hormone does not require secondary messenger for their action?

(1) Gonadotropins

(2) Growth hormone

(3) Epinephrine

(4) Triiodothyronine

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154. Read the following statement and choose the correct statement ?

(1) Hormone are nutrient chemical which act as intercellular messenger.

(2) Each receptor is specific to one hormone only.

(3) Steriod hormone mostly regulate gene expression or chromosome function by the interaction of hormone receptor complex on the membrane.

(4) All hormone are insoluble in water.

155. Read the following column-I & II and identify their correct match :-

	Column-I		Column-II
(A)	Growth hormone	(i)	Pancreas
(B)	Somatostatin	(ii)	Adrenal medulla
(C)	Catecholemines	(iii)	Pineal gland
(D)	Melatonin	(iv)	Pituitary gland
		(v)	Hypothalamus
		(vi)	Adrenal cortex

(1) A-iv, B-v, C-ii, D-iii

(2) A-v, B-iv, C-vi, D-i

(3) A-ii, B-iii, C-iv, D-vi

(4) A-iii, B-i, C-ii, D-iv

156. **Statement-I** : Prolonged hyperglycemia leads to a complex disorder diabetes mellitus.

Statement-II : Glucagon is a hypoglycemic hormone while insulin is a hyperglycemic hormone.

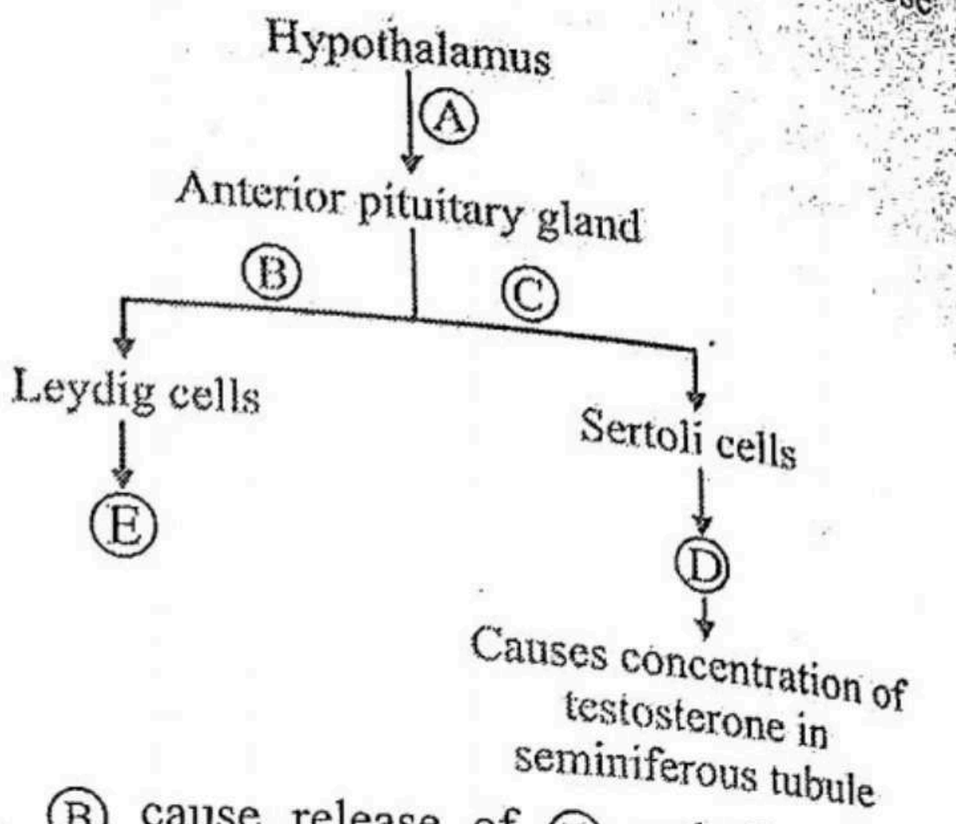
(1) Statement-I and II both are correct.

(2) Statement-I and II both are incorrect.

(3) Statement-I is correct but II is incorrect.

(4) Statement-II is correct but I is incorrect.

157. Observe the following flow chart and choose right statement :-



(1) (B) cause release of (E) and (E) inhibits secretion of (D) (normally)

(2) (C) causes release of (D) and (D) concentrates (E) inside the seminiferous tubule

(3) (A) is GnRH, (C) is ICSH and (B) is FSH

(4) (A) is GnRH, (C) is LH and (B) is FSH

158. Read the following statements :-

(a) Hormones are nutrient chemicals.

(b) Pancreas is organized endocrine gland.

(c) Anterior pituitary is under direct neural regulation of hypothalamus.

(d) Growth hormone is secreted by anterior pituitary.

(e) Deficiency of vasopressin cause acromegaly.

Choose the correct options :-

(1) a,b, & c (2) b,c & d

(3) a,c & d (4) b & d

159. Which of the following terms are "not" related with amoeboid movement ?

(1) Pseudopodia (2) Macrophages

(3) Flagella (4) Neutrophils

160. In the centre of I-Band is an elastic fiber present, which bisect I-band, is called as :

(1) 'H' -zone (2) 'M' -line

(3) 'Z' line (4) 'A' -band

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161. In the resting state a subunit of troponin masks the active binding sites for myosin on :

(1) Actin filaments (2) Meromyosin filament

(3) M–line (4) H–zone

162. A sarcomere consists of

(1) One A-band and two I-band

(2) Half A-band and two half I-band

(3) Half A-band and one I-band

(4) One A-band and two half I-band

163. Match the column I & II with correct option.

	Column-I		Column-II
(A)	Sarcomere	(i)	Red muscle fiber
(B)	Actin filament	(ii)	White muscle fiber
(C)	Slow muscle fiber	(iii)	Contractile unit
(D)	Fast muscle fiber	(iv)	I-band

(1) A-iii, B-iv, C-i, D-ii

(2) A-i, B-ii, C-iii, D-iv

(3) A-iv, B-i, C-iii, D-ii

(4) A-ii, B-iii, C-iv, D-i

164. Select the correct option :-

(1) First vertebra is the axis and it articulates with the occipital condyles.

(2) Sternum is a flat bone on the dorsal midline of thorax.

(3) Thoracic vertebrae, ribs and sternum together form the rib cage.

(4) 11th and 12th pairs of ribs are called vertebrochondral ribs.

165. At the base of skull, a large foramen is present which allows brain to continue with the spinal cord. That foramen is called :

(1) Foramen ovale (2) Foramen magnum

(3) Vertebral foramen (4) Obturator foramen

166. A single unpaired 'U' shaped bone present in the neck region :

(1) Vomer (2) Hyoid

(3) Mandible (4) Clavicle

167. Which of the following is a synovial joint :

(1) Ball and socket joint

(2) Hinge joint

(3) Fibrous joint

(4) Both 1 & 2

168. Which of the following not the constituent of axial skeleton :

(1) Skull (2) Patella

(3) Ribs (4) Sternum

169. Schwann cell are found in :-

(1) Myelinated nerve fibres of PNS

(2) Unmyelinated nerve fibres of PNS

(3) Both 1 and 2

(4) Myelinated nerve fibers of CNS

170. The components of the human brain responsible for regulating sexual behaviour, excitement, pleasure, rage, and fear are:

(1) Pons & medulla oblongata

(2) Brain stem & Thalamus

(3) Corpus callosum & Motor area

(4) Limbic system & hypothalamus

171. Which of the following structure contains centres, which control respiration, cardiovascular reflexes and gastric secretions:-

(1) Mid brain

(2) Cerebellum

(3) Pons

(4) Medulla oblongata

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172. Read column-I, II & III and identify their correct match :-

	Column-I		Column-II		Column-III
(A)	Aldosterone	(P)	Alpha cells	(i)	Increases blood glucose levels.
(B)	Epinephrine	(Q)	Beta cells	(ii)	Decreases blood glucose level
(C)	Insulin	(R)	Adrenal medulla	(iii)	Reabsorption of sodium
(D)	Glucagon	(S)	Adrenal cortex	(iv)	Increases heat beat

(1) A-P-i, B-Q-ii, C-R-iii, D-S-iv

(2) A-Q-i, B-P-ii, C-S-iii, D-R-iv

(3) A-Q-i, B-P-iv, C-S-iii, D-R-ii

(4) A-S-iii, B-R-iv, C-Q-ii, D-P-i

173. Progesteron hormone ?

(1) Stimulates testicular development

(2) Inhibits fetal development

(3) Stimulates formation of milk alveoli

(4) Stimulates secondary sexual character

174. Nor epinephrine is secreted by :-

(1) Adrenal medulla and increases the heart rate

(2) Adrenal medulla and decreases the heart rate

(3) Adrenal cortex and increases the heart rate

(4) Adrenal cortex and decreases the heart rate

175. The secretion of PTH is regulated by :-

(1) Releasing hormone secreted by anterior pitutary

(2) Vit-D_3

(3) Circulating Ca^{++} ion level

(4) All of the above

76. Which of the following hormones controls the functioning of adrenal cortex?

(1) TSH (2) FSH (3) ACTH (4) LH

177. Select the correct statements :-

(A) Exophthalmic goitre is a form of hypothyroidism

(B) Hypothyroidism may cause menstrual cycle to become irregular.

(C) MSH acts on the melanocytes and regulates pigmentation of the skin.

(D) Adenohypophysis consists of two portions, pars distalis and pars nervosa.

(1) B, D (2) B, C (3) A, B (4) A, D

178. Which of the following disorder affects the neuromuscular junction (NMJ) leading to fatigue, weakening and paralysis of skeletal muscle?

(1) Myasthenia gravis (2) Muscular dystrophy

(3) Osteoporosis (4) Tetany

179. Match List-I with List-II :-

	List-I		List-II
(A)	Scapula	(I)	Acromian
(B)	Floating ribs	(II)	11th and 12th pairs
(C)	Vertebro sternal ribs	(III)	1 to 7th pairs
(D)	Pectoral girdls	(IV)	Scapula, clavicle

Choose the correct answer from the options given below :-

(1) A-III, B-IV, C-II, B-I

(2) A-II, B-IV, C-III, D-I

(3) A-I, B-IV, C-II, D-III

(4) A-I, B-II, C-III, B-IV

180. Match the List I to List II :

List-I		List-II	
A	Myosin	i	Globular protein
B	Actin	ii	Sarcomere
C	Two adjacent	iii	Meromyosin
D	Smooth mussels	iv	Fallopian tube

(1) A - iii, B - i, C - ii, D - iv

(2) A - iii, B - ii, C - i, D - iv

(3) A - ii, B - iii, C - i, D - iv

(4) A - iv, B - iii, C - ii, D - i

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TEST # 03

ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Ans.	1	2	1	3	1	2	1	3	2	3	1	1	3	4	1	2	3	2	2	4
Que.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
Ans.	1	1	2	1	1	3	3	3	2	1	2	4	1	4	1	1	2	4	4	3
Que.	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Ans.	3	2	3	2	4	3	1	2	2	3	3	2	1	3	1	3	1	4	2	4
Que.	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80
Ans.	2	2	3	1	1	1	4	4	3	3	4	1	4	3	3	2	4	3	1	3
Que.	81	82	83	84	85	86	87	88	89	90	91	92	93	94	95	96	97	98	99	100
Ans.	3	4	2	2	3	1	2	2	3	4	4	3	2	4	4	1	1	2	4	4
Que.	101	102	103	104	105	106	107	108	109	110	111	112	113	114	115	116	117	118	119	120
Ans.	3	2	4	2	2	4	2	3	4	2	1	2	1	3	3	3	1	3	2	3
Que.	121	122	123	124	125	126	127	128	129	130	131	132	133	134	135	136	137	138	139	140
Ans.	3	4	4	1	4	1	2	4	1	3	2	4	3	3	2	3	2	4	3	3
Que.	141	142	143	144	145	146	147	148	149	150	151	152	153	154	155	156	157	158	159	160
Ans.	1	3	3	1	2	2	4	2	4	2	4	4	4	2	1	3	2	4	3	3
Que.	161	162	163	164	165	166	167	168	169	170	171	172	173	174	175	176	177	178	179	180
Ans.	1	4	1	3	2	2	4	2	3	4	4	4	3	1	3	3	2	1	4	1

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